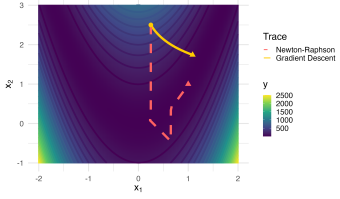
Optimization in Machine Learning

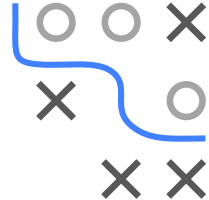
Second order methods
Fisher Scoring



Learning goals

- Fisher Scoring
- Newton-Raphson vs. Fisher scoring
- Logistic regression

RECAP OF NEWTON'S METHOD



- Second-order Taylor expansion of log-likelihood around the current iterate $\boldsymbol{\theta}^{[t]}$:

$$\ell(\boldsymbol{\theta}) \approx \ell(\boldsymbol{\theta}^{[t]}) + \nabla \ell(\boldsymbol{\theta}^{[t]})^\top (\boldsymbol{\theta} - \boldsymbol{\theta}^{[t]}) + \frac{1}{2}(\boldsymbol{\theta} - \boldsymbol{\theta}^{[t]})^\top [\nabla^2 \ell(\boldsymbol{\theta}^{[t]})](\boldsymbol{\theta} - \boldsymbol{\theta}^{[t]})$$

- We then differentiate w.r.t. $\boldsymbol{\theta}$ and set the gradient to zero:

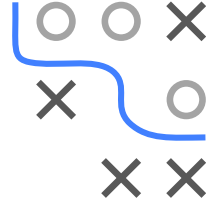
$$\nabla \ell(\boldsymbol{\theta}^{[t]}) + [\nabla^2 \ell(\boldsymbol{\theta}^{[t]})](\boldsymbol{\theta} - \boldsymbol{\theta}^{[t]}) = \mathbf{0}$$

- Solving for $\boldsymbol{\theta}^{(t)}$ yields the pure Newton-Raphson update:

$$\boldsymbol{\theta}^{[t+1]} = \boldsymbol{\theta}^{[t]} + [-\nabla^2 \ell(\boldsymbol{\theta}^{[t]})]^{-1} \nabla \ell(\boldsymbol{\theta}^{[t]})$$

- **Potential stability issue**: pure Newton-Raphson updates do not always converge. Its quadratic convergence rate is "local" in the sense that it requires starting close to a solution.

FISHER SCORING



- Fisher's scoring method replaces the negative *observed Hessian* $-\nabla^2\ell(\boldsymbol{\theta})$ by the Fisher information matrix, i.e., the variance of $\nabla\ell(\boldsymbol{\theta})$, which, under weak regularity conditions, equals the negative *expected Hessian*

$$\mathbb{E}[\nabla\ell(\boldsymbol{\theta})\nabla\ell(\boldsymbol{\theta})^\top] = \mathbb{E}[-\nabla^2\ell(\boldsymbol{\theta})],$$

  and is positive semi-definite under exchangeability of expectation and differentiation
- **NB**: it can be shown that $\mathbb{E}[\nabla\ell(\boldsymbol{\theta})] = \mathbf{0}$, which provides the expression of the variance of $\nabla\ell(\boldsymbol{\theta})$ as the expected outer product of the gradients
- Therefore the Fisher scoring iterates are given by

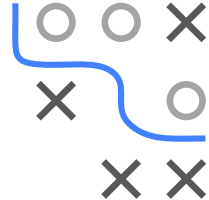
$$\boldsymbol{\theta}^{[t+1]} = \boldsymbol{\theta}^{[t]} + \mathbb{E}[-\nabla^2\ell(\boldsymbol{\theta}^{[t]})]^{-1}\nabla\ell(\boldsymbol{\theta}^{[t]})$$

NEWTON-RAPHSON VS. FISHER SCORING

Aspect	Newton-Raphson	Fisher scoring
Second-order Matrix	Exact negative Hessian matrix	Fisher information matrix
Curvature	Exact	Approximated
Computational Cost	Higher	Lower (often has a simpler structure)
Convergence	Fast but potentially unstable	Slower but more stable
Positive Definite	Not guaranteed	Yes with Fisher information
Use Case	General non-linear optimization	Likelihood-based models, especially GLMs

- In many cases Newton-Raphson and Fisher scoring are equivalent (see below)

LOGISTIC REGRESSION



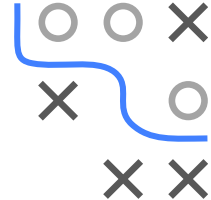
- The goal of logistic regression is to predict a binary event
- Given n observations $\left(\boldsymbol{x}^{(i)}, y^{(i)}\right) \in \mathbb{R}^{p+1} \times \{0, 1\}$, $y^{(i)} | \, \boldsymbol{x}^{(i)} \sim \textit{Bernoulli}(\pi^{(i)})$
- We want to minimize the following risk

$$\mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) \quad = \quad -\sum_{i=1}^{n} y^{(i)} \log(\pi^{(i)}) + \left(1 - y^{(i)}\right) \log(1 - \pi^{(i)})$$

with respect to $\boldsymbol{\theta}$, where the probabilistic classifier $\pi^{(i)} = \pi\left(\boldsymbol{x}^{(i)} \mid \boldsymbol{\theta}\right) = s\left(f\left(\boldsymbol{x}^{(i)} \mid \boldsymbol{\theta}\right)\right)$, the sigmoid function $s(f) = \frac{1}{1+\exp(-f)}$ and the score $f\left(\boldsymbol{x}^{(i)} \mid \boldsymbol{\theta}\right) = \boldsymbol{\theta}^\top \boldsymbol{x}$

- **NB**: Note that $\frac{\partial}{\partial f} s(f) = s(f)(1 - s(f))$ and $\frac{\partial f\left(\boldsymbol{x}^{(i)} \mid \boldsymbol{\theta}\right)}{\partial \boldsymbol{\theta}} = \left(\boldsymbol{x}^{(i)}\right)^\top$
- For more details we refer to the i2ml lecture

LOGISTIC REGRESSION



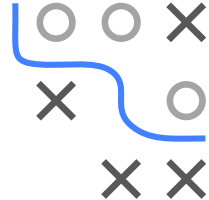
- Partial derivative of empirical risk using chain rule:

$$
\begin{aligned}
\frac{\partial}{\partial \boldsymbol{\theta}} \mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) &= -\sum_{i=1}^{n} \frac{\partial}{\partial \pi^{(i)}} \left(y^{(i)} \log(\pi^{(i)}) + (1 - y^{(i)}) \log(1 - \pi^{(i)})\right) \frac{\partial \pi^{(i)}}{\partial \boldsymbol{\theta}} \\
&= -\sum_{i=1}^{n} \left(\frac{y^{(i)}}{\pi^{(i)}} - \frac{1 - y^{(i)}}{1 - \pi^{(i)}}\right) \frac{\partial s(f(\boldsymbol{x}^{(i)} \mid \boldsymbol{\theta}))}{\partial f(\boldsymbol{x}^{(i)} \mid \boldsymbol{\theta})} \frac{\partial f(\boldsymbol{x}^{(i)} \mid \boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \\
&= \sum_{i=1}^{n} \left(\pi^{(i)} - y^{(i)}\right) \left(\boldsymbol{x}^{(i)}\right)^{\top} \\
&= (\pi(\mathbf{X} \mid \boldsymbol{\theta}) - \boldsymbol{y})^{\top} \mathbf{X}
\end{aligned}
$$

where $\mathbf{X} = \left(\boldsymbol{x}^{(1)\top}, \ldots, \boldsymbol{x}^{(n)\top}\right)^{\top} \in \mathbb{R}^{n \times (p+1)}$, $\boldsymbol{y} = \left(y^{(1)}, \ldots, y^{(n)}\right)^{\top}$,
$\pi(\mathbf{X} \mid \boldsymbol{\theta}) = \left(\pi^{(1)}, \ldots, \pi^{(n)}\right)^{\top} \in \mathbb{R}^{n}$

- $\nabla_{\boldsymbol{\theta}} \mathcal{R}_{\text{emp}} = \left(\frac{\partial}{\partial \boldsymbol{\theta}} \mathcal{R}_{\text{emp}}\right)^{\top}$

LOGISTIC REGRESSION



- The Hessian of logistic regression:

$$\nabla^2_{\boldsymbol{\theta}} \mathcal{R}_{\text{emp}} = \frac{\partial^2}{\partial \boldsymbol{\theta}^\top \partial \boldsymbol{\theta}} \mathcal{R}_{\text{emp}} = \frac{\partial}{\partial \boldsymbol{\theta}^\top} \sum_{i=1}^{n} \left(\pi^{(i)} - y^{(i)}\right) \left(\boldsymbol{x}^{(i)}\right)^\top$$

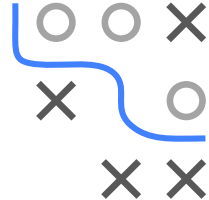
$$= \sum_{i=1}^{n} \boldsymbol{x}^{(i)} \left(\pi^{(i)} \left(1 - \pi^{(i)}\right)\right) \left(\boldsymbol{x}^{(i)}\right)^\top$$

$$= \mathbf{X}^\top \mathbf{D} \mathbf{X}$$

where $\mathbf{D} \in \mathbb{R}^{n \times n}$ is a diagonal matrix containing the variances of $y^{(i)}$ on the diagonals

$$\mathbf{D} = \text{diag}\left(\pi^{(1)}(1 - \pi^{(1)}), \ldots, \pi^{(n)}(1 - \pi^{(n)})\right)$$

LOGISTIC REGRESSION



- We now have

$$\nabla_{\boldsymbol{\theta}} \mathcal{R}_{\text{emp}} = \mathbf{X}^\top \left(\pi(\mathbf{X}|\, \boldsymbol{\theta}) - \boldsymbol{y}\right)$$

$$\nabla^2_{\boldsymbol{\theta}} \mathcal{R}_{\text{emp}} = \mathbf{X}^\top \mathbf{D} \mathbf{X}$$

- Newton-Raphson:

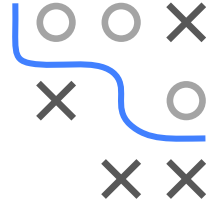
$$\boldsymbol{\theta}^{[t+1]} = \boldsymbol{\theta}^{[t]} - [\mathbf{X}^\top \mathbf{D} \mathbf{X}]^{-1} \nabla_{\boldsymbol{\theta}^{[t]}} \mathcal{R}_{\text{emp}}$$

- Fisher scoring:

$$\boldsymbol{\theta}^{[t+1]} = \boldsymbol{\theta}^{[t]} - \mathbb{E}[\mathbf{X}^\top \mathbf{D} \mathbf{X}]^{-1} \nabla_{\boldsymbol{\theta}^{[t]}} \mathcal{R}_{\text{emp}}$$

- Note that the Hessian does not depend on the $y^{(i)}$ explicitly but only depends on $\mathbb{E}[y^{(i)}] = \pi^{(i)}$. Thus the expectation of the observed Hessian w.r.t. $y^{(i)} \sim P(y^{(i)}|\boldsymbol{x}^{(i)}, \boldsymbol{\theta})$ coincides with $\nabla^2_{\boldsymbol{\theta}} \mathcal{R}_{\text{emp}}(\boldsymbol{\theta})$ itself

GENERALIZED LINEAR MODELS



- $y|\boldsymbol{x}$ belongs to an **exponential family** with density:

$$p(y|\delta,\phi) = exp\left\{\frac{y\delta - b(\delta)}{a(\phi)} + c(y,\phi)\right\},$$

  where δ is the natural parameter and $\phi > 0$ is the dispersion parameter

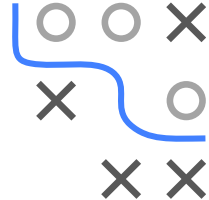
- We often take $a_i(\phi) = \frac{\phi}{w_i}$, with ϕ a positive constant, and w_i is a weight

- Generalized linear models (GLMs) relate the conditional mean $\mu(\boldsymbol{x}) = \mathbb{E}[y|\boldsymbol{x}]$ of y to a linear predictor η via a strictly increasing link function $g(\mu) = \eta = \boldsymbol{x}^\top\theta$

- One can show that mean $\mu = \mu(\boldsymbol{x}) = b'(\delta) = g^{-1}(\eta)$, variance $Var(y|\boldsymbol{x}) = a(\phi)b''(\delta)$, where

$$\frac{\partial b(\delta)}{\partial\theta} = \frac{\partial b(\delta)}{\partial\delta}\frac{\partial\delta}{\partial\mu}\frac{\partial\mu}{\partial\eta}\frac{\partial\eta}{\partial\theta} = \mu\frac{1}{b''(\delta)}\frac{1}{g'(\mu)}\boldsymbol{x}$$

GENERALIZED LINEAR MODELS



- We can estimate δ using MLE with sample $(\boldsymbol{x}^{(i)}, y^{(i)})$ for $i = 1, \ldots, n$
- Take $a^{(i)}(\phi) = \frac{\phi}{w^{(i)}}$, ϕ is a positive constant, we could ignore it since the goal is to maximize the function:

$$
\begin{aligned}
\nabla \ell_\theta(\delta, \phi) &= \sum_{i=1}^{n} \frac{w_i(y^{(i)} - \mu^{(i)})}{b''(\delta) g'(\mu^{(i)})} \boldsymbol{x}^{(i)} \\
&= \sum_{i=1}^{n} \frac{w^{(i)}(y^{(i)} - \mu^{(i)}) g'(\mu^{(i)})}{b''(\delta)[g'(\mu^{(i)})]^2} \boldsymbol{x}^{(i)} \\
&= \mathbf{X}^\top \mathbf{W} \mathbf{G} (\mathbf{Y} - \boldsymbol{\mu})
\end{aligned}
$$

- $\mathbf{W}$ is a diagonal matrix with element $\frac{w^{(i)}}{b''(\delta)[g'(\mu^{(i)})]^2}$
- $\mathbf{G}$ is a diagonal matrix with element $g'(\mu^{(i)})$

GENERALIZED LINEAR MODELS

$$
\begin{aligned}
-\nabla^2 \ell_\theta(\delta, \phi) = &\sum_{i=1}^{n} \frac{w^{(i)}}{b''(\delta)[g'(\mu^{(i)})]^2} \boldsymbol{x}^{(i)} \boldsymbol{x}^{(i)\top} \\
&+ \sum_{i=1}^{n} \frac{w^{(i)}(y^{(i)} - \mu^{(i)})(b''(\delta) g''(\mu^{(i)}) / g'(\mu^{(i)}))}{[b''(\delta) g'(\mu^{(i)})]^2} \boldsymbol{x}^{(i)} \boldsymbol{x}^{(i)\top} \\
&+ \sum_{i=1}^{n} \frac{w^{(i)}(y^{(i)} - \mu^{(i)})(b'''(\delta) / b''(\delta))}{[b''(\delta) g'(\mu^{(i)})]^2} \boldsymbol{x}^{(i)} \boldsymbol{x}^{(i)\top}
\end{aligned}
$$

$$
\mathbb{E}[-\nabla^2 \ell_\theta(\delta, \phi)] = \sum_{i=1}^{n} \frac{w^{(i)}}{b''(\delta)[g'(\mu^{(i)})]^2} \boldsymbol{x}^{(i)} \boldsymbol{x}^{(i)\top} = \mathbf{X}^\top \mathbf{W} \mathbf{X}
$$

- Iteratively Reweighted Least Squares (IRLS) with weights $\frac{w^{(i)}}{b''(\delta)[g'(\mu^{(i)})]^2}$

GENERALIZED LINEAR MODELS

- Fisher scoring:

$$\begin{aligned}\boldsymbol{\theta}^{[t+1]} &= \boldsymbol{\theta}^{[t]} + (\mathbf{X}^\top \mathbf{W}\mathbf{X})^{-1}\mathbf{X}^\top \mathbf{W}\mathbf{G}(\mathbf{Y} - \boldsymbol{\mu}) \\ &= (\mathbf{X}^\top \mathbf{W}\mathbf{X})^{-1}\mathbf{X}^\top \mathbf{W}\left(\mathbf{G}(\mathbf{Y} - \boldsymbol{\mu}) + \mathbf{X}\boldsymbol{\theta}^{[t]}\right)\end{aligned}$$

- For canonical link where $\eta = \delta$ $(= g(\mu) = \boldsymbol{x}^\top \theta)$, the second and third term of Hessian cancel each other out and Hessian coincides with Fisher information matrix since

$$\frac{\partial \eta}{\partial \delta} = 1 \Rightarrow b''(\delta) = \frac{1}{g'(\mu^{(i)})} \Rightarrow \frac{b'''(\delta)}{b''(\delta)} = -\frac{g''(\mu^{(i)})}{[g'(\mu^{(i)})]^2}$$

- This will now be a convex problem with Fisher scoring equal to Newton's method

- There are also hybrid algorithms that start out with IRLS which is easier to initialize, and switch over to Newton-Raphson after some iterations